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Exact Computation for the Normalized Power Prior

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Master's Thesis presented to the Graduate Program in Applied Mathematics and Data Science of the School of Applied Mathematics as a requirement to obtain the degree of Master in Applied Mathematics and Data Science.

School of Applied Mathematics
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Statement of Authorship

I hereby declare that the thesis submitted is my own work. All direct or indirect sources used are acknowledged as references. I further declare that I have not submitted this thesis at any other institution in order to obtain a degree.

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I would like to thank [many people]

Abstract

In statistical modelling with historical data, a central challenge is how to effectively integrate prior evidence with current observations. This issue is particularly pressing in clinical trials and medical applications, where using historical data can improve both the efficiency and robustness of inference. A natural Bayesian strategy for this purpose is the use of informative priors. Among them, the normalised power prior has attracted significant interest, as it provides a principled way to control the influence of historical information. The method introduces a power parameter, $\eta \in [0, 1]$, that governs the degree of borrowing: $\eta = 0$ discards the past data while $\eta = 1$ fully incorporates it. This flexibility makes the framework highly appealing in practice. Despite its advantages, the power prior posterior distribution is doubly intractable, as it requires evaluation of a normalising constant that depends on η. Existing approaches rely on approximate inference, which often lack theoretical guarantees and explicit convergence properties. To overcome these limitations, we propose an exact MCMC algorithm based on pseudo-marginal methods. The key idea is to replace the intractable ratio of normalising constants with an unbiased estimator. After finding useful identities for the derivative of the log normalising constant, we construct unbiased estimators of log-ratios through Monte Carlo integration, enabling exact acceptance probabilities, by employing the Poisson Estimator Method, within a Metropolis-Hastings or Barker's scheme. We evaluate the method on Gaussian models under varying degrees of compatibility between historical and current data. Results demonstrate that the algorithm accurately reflects data conflict by shrinking η when the datasets are incompatible, while recovering correct posterior behaviour when they are aligned. Comparisons with analytical baselines confirm that our method achieves reliable estimates of both model parameters and the power parameter, with efficiency suitable for practical use. This provides a rigorous, exact alternative to approximate approaches, offering stronger guarantees for applications such as adaptive clinical trial design.

Keywords: Bayesian Inference; Power Prior; Doubly Intractable Distributions; Exact MCMC; Historical Data Analysis

Resumo

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1 Introduction

Luiz: Second-to-last thing to be written. Don't sweat it. Leave it to *Papa*

- Describe NPP and notation; motivation with historical data (MANY CITATIONS).
- The NPP as a doubly-intractable problem (the prior is intractable)
- Mention that we start with approximate methods and then move on to exact stuff

In statistical modelling with historical data, a central challenge is effectively integrating information from past studies with current data. This is particularly relevant in clinical trials and medical research, where leveraging historical data can improve the efficiency and robustness of statistical inferences. A natural approach in this setting is the use of *informative priors*, which incorporate prior knowledge into the analysis, thereby enhancing parameter estimation and decision-making. Adame: Some references here would be nice.

Among the various methods for constructing informative priors, *power priors* (Ibrahim et al., 2015) have gained significant attention. They allow for controlled borrowing of information from historical data by adjusting a power parameter, η, which typically ranges between 0 and 1. When $\eta = 0$, no historical information is incorporated, whereas $\eta = 1$ fully integrates the historical data into the current analysis. Adame: I would like to introduce the NPPs issues here and then talk about the motivation for the exact inference. Didn't find a good way to do it.

Despite their advantages, the posterior distribution in these models is often *doubly intractable*, making standard Markov Chain Monte Carlo (MCMC) methods challenging to apply. Current approaches rely on *approximate methods* (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021), which lack theoretical guarantees and explicit convergence bounds. This work addresses this gap by developing an *exact MCMC algorithm* (Bhattacharya, 2021; Murray; Ghahramani; MacKay, 2012) capable of efficiently sampling from these complex posterior distributions.

2 Background

2.1 Bayesian Inference

2.1.1 Principles

Bayesian inference provides a probabilistic framework for updating uncertainty about unknown quantities after observing data.

Let D denote the observed data, and let $\theta \in \Theta$ denote an unknown parameter defined on a measurable parameter space $(\Theta, \mathcal{F})$. The Bayesian model is specified by a likelihood function $L(D \mid \theta)$ and a prior distribution $\pi_0(\theta)$.

The likelihood describes the information about θ contained in the data, while the prior represents information available before observing D. Bayes' theorem combines these two sources of information through the posterior distribution

$$\pi(\theta \mid D) = \frac{L(D \mid \theta)\pi_0(\theta)}{\int_\Theta L(D \mid t)\pi_0(t)\,\mathrm{d}t}. \tag{2.1}$$

The denominator in (2.1) is the marginal likelihood, or evidence, and is given by

$$m(D) = \int_\Theta L(D \mid t)\pi_0(t)\,\mathrm{d}t. \tag{2.2}$$

When $m(D) < \infty$, the posterior distribution is well-defined and integrates to one. In this sense, Bayesian inference can be understood as a normalization procedure that transforms the unnormalized density $L(D \mid \theta)\pi_0(\theta)$ into a probability distribution over Θ.

For any posterior-integrable function $h : \Theta \to \mathbb{R}$, posterior summaries are often expressed through expectations of the form

$$\mathbb{E}_{\pi(\theta|D)}[h(\theta)] = \int_\Theta h(\theta)\pi(\theta \mid D)\,\mathrm{d}\theta. \tag{2.3}$$

Examples include posterior means, posterior variances, marginal probabilities, credible intervals, and posterior predictive quantities. The main computational challenge is that the posterior distribution is usually available only up to a normalizing constant.

Indeed, many Bayesian algorithms only require the unnormalized posterior density

$$\gamma(\theta) = L(D \mid \theta)\pi_0(\theta), \tag{2.4}$$

because the marginal likelihood $m(D)$ does not depend on θ.

However, the marginal likelihood is still important for model comparison, empirical Bayes procedures, and doubly intractable models. In low-dimensional conjugate models, the integral in (2.2) may be available analytically.

In modern Bayesian models, this integral is often unavailable, and posterior expectations such as (2.3) must be approximated numerically. This motivates the use of simulation-based methods.

2.1.2 Monte Carlo Methods

Monte Carlo methods approximate integrals by random sampling.

Let π be a probability distribution on $(\Theta, \mathcal{F})$, and suppose that the quantity of interest is

$$\mu = \mathbb{E}_\pi[h(\theta)] = \int_\Theta h(\theta)\pi(\theta)\,\mathrm{d}\theta. \tag{2.5}$$

If $\theta_1, \ldots, \theta_N$ are independent samples from π, then the Monte Carlo estimator of μ is

$$\widehat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} h(\theta_i). \tag{2.6}$$

Under standard integrability conditions, the law of large numbers gives

$$\widehat{\mu}_N \xrightarrow[N\to\infty]{a.s.} \mu. \tag{2.7}$$

If $h(\theta)$ has finite variance under π, the central limit theorem gives

$$\sqrt{N}(\widehat{\mu}_N - \mu) \xrightarrow[N\to\infty]{d} \mathcal{N}\Big(0, \sigma_h^2\Big), \tag{2.8}$$

where

$$\sigma_h^2 = \mathrm{Var}_\pi\,[h(\theta)]. \tag{2.9}$$

The important feature of the Monte Carlo approximation is that its convergence rate is typically $N^{-1/2}$ and does not directly depend on the dimension of Θ. This makes Monte Carlo methods attractive for Bayesian inference, where posterior distributions are often high-dimensional.

The difficulty is that independent sampling from $\pi(\theta \mid D)$ is rarely possible. Markov chain Monte Carlo methods address this difficulty by constructing a Markov chain whose invariant distribution is the desired posterior distribution.

The samples are then dependent, but posterior expectations can still be estimated using ergodic averages. If $(\theta_n)_{n\geq 0}$ is a Markov chain with invariant distribution π, then the MCMC estimator is

$$\widehat{\mu}_N^{\mathrm{MCMC}} = \frac{1}{N}\sum_{n=1}^{N} h(\theta_n). \tag{2.10}$$

Under suitable irreducibility, recurrence, and integrability conditions, the ergodic theorem gives

$$\widehat{\mu}_N^{\mathrm{MCMC}} \xrightarrow[N\to\infty]{a.s.} \mathbb{E}_\pi[h(\theta)]. \tag{2.11}$$

The price paid for replacing independent samples with a Markov chain is autocorrelation.

When samples are correlated, the effective amount of information in N iterations may be much smaller than N independent draws. This motivates the use of diagnostics such as effective sample size, Monte Carlo standard errors, trace plots, and comparisons across independent chains. Another important practical issue is initialization.

A Markov chain usually starts from a distribution different from π, and therefore the early iterations may not be representative of the target distribution. The usual practical response is to discard an initial portion of the chain, known as burn-in.

However, burn-in is an asymptotic correction rather than an exact one. This distinction is central to the later discussion of exact and unbiased MCMC methods.

2.2 Standard MCMC Methods

`Adame:` Since I know you will ask me what do I mean by standard: we can surely think of a better name

2.2.1 Gibbs Sampling

2.2.2 Metropolis-Hastings

2.3 Doubly Intractable Distributions

2.4 The Normalized Power Prior

2.4.1 Formulation

Consider the following Bayesian schema:

$$p(\theta \mid D) = \frac{L(D \mid \theta)\pi_0(\theta)}{\int_\Theta L(D \mid t)\pi(t)\,\mathrm{d}t} = \frac{L(D \mid \theta) \quad \overbrace{\pi_0(\theta)}^{\text{Initial prior}}}{m(D)}. \tag{2.12}$$

Definição 2.4.1 (Power prior)**.** *Let $D_0 = \{d_{01}, d_{02}, \ldots, d_{0N_0}\}$, $d_{0i} \subseteq \mathcal{X}^p$ be the historical data and let $L_0(D_0 \mid \theta)$ be a likelihood function assumed to be finite for all $\theta \in \Theta \subseteq \mathbb{R}^p$. Then, the power prior is defined as*

$$\pi_\eta(\theta \mid D_0) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta), \tag{2.13}$$

where $\pi_0(\theta)$ is the initial prior distribution of θ and η is the (fixed) power parameter.

Then, when we want to accomodate the uncertainty in the choice of η, we treat it as a random variable and assign a prior distribution to it, indexed by φ. In this case, we find the *unnormalized* power prior as

$$\pi^\star(\theta, \eta \mid D_0, \varphi) = \pi_\eta(\theta \mid D_0)\pi_A(\eta \mid \varphi) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta)\pi_A(\eta \mid \varphi). \tag{2.14}$$

In order to compute the posterior distribution of θ and η, we need to find the normalizing constant (from (2.13)) $c_0(\eta)$, which is given by

$$c_0(\eta) = \int_\Theta L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t) \tag{2.15}$$

and the marginal posterior distribution of η as, for a new dataset D,

$$p(\eta \mid D, D_0, \varphi) \propto \frac{\pi_A(\eta \mid \varphi)}{c_0(\eta)} \int_\Theta L(D \mid t)L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t). \tag{2.16}$$

Notice that, since both $c_0(\eta)$ and $\int_\Theta L(D \mid t)L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t)$ are intractable, we call this a doubly-intractable posterior. Previous work (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021) have focused on emulating $\log c_0(\eta)$ and $\log c'(\eta) = \mathbb{E}_{\pi_\eta}[\log L_0(D_0 \mid \theta)]$ (2.18) through non-parametric models, which has proven to be more effective than common interpolation methods—particularly when incorporating shape constraints.

Sampling from these posteriors is challenging due to the intractability of standard M-H acceptance probabilities, making approximate methods like auxiliary variable techniques computationally expensive and lacking convergence guarantees; an alternative is unbiased event constructions, which enable exact sampling without evaluating normalizing constants.

For instance, applying Barker's algorithm for MCMC (Vats et al., 2021) to sample from a target distribution $\pi(x) \propto \pi'(x)$ with proposal density $q(x \mid y)$ has become feasible through Bernoulli factory methods. These methods enable the construction of events with probability $\alpha_B(x, y)$ when

$$\alpha_B(x,y) = \frac{\pi(y)q(y \mid x)}{\pi(x)q(x \mid y) + \pi(y)q(y \mid x)} = \frac{\pi'(y)q(y \mid x)}{\pi'(x)q(x \mid y) + \pi'(y)q(y \mid x)} \tag{2.17}$$

cannot be directly evaluated (Gonçalves; Łatuszyński; Roberts, 2017).

2.4.2 Properties

2.4.2.1 Derivative of the normalizing constant

Considering $c_0(\eta)$ as in (2.15), we can find the derivative of $c_0(\eta)$ with respect to η as

$$\begin{aligned}
\frac{\mathrm{d}c_0(\eta)}{\mathrm{d}\eta} &= \frac{\mathrm{d}}{\mathrm{d}\eta}\int_\Theta L(D_0 \mid t)^\eta \,\mathrm{d}P_0(t), \\
&= \int_\Theta \frac{\mathrm{d}}{\mathrm{d}\eta} L(D_0 \mid t)^\eta \,\mathrm{d}P_0(t), \\
&= \int_\Theta \log(L(D_0 \mid t)) L(D_0 \mid t)^\eta \,\mathrm{d}P_0(t), \\
&= \int_\Theta \log(L(D_0 \mid t)) \pi_\eta(t \mid D_0) \,\mathrm{d}P_0(t), \\
&= c_0(\eta)\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))],
\end{aligned}$$

where $\mathbb{E}_{\pi_\eta}[\cdot]$ denotes the expectation with respect to the power prior distribution $\pi_\eta(\theta \mid D_0)$.

When we're interested on the derivative of the logarithm of the normalizing constant, we can use the following expression:

$$\frac{\mathrm{d}\log c_0(\eta)}{\mathrm{d}\eta} = \frac{1}{c_0(\eta)}\frac{\mathrm{d}c_0(\eta)}{\mathrm{d}\eta} = \mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))]. \tag{2.18}$$

2.4.3 Other Historical Data Modelling Approaches

2.5 Unbiased MCMC Methods

2.5.1 Barker's Algorithm for Exact Inference

Barker's algorithm (Barker, 1965) provides an alternative to Metropolis-Hastings that is amenable to Bernoulli factory methods. For a target distribution $\pi(\theta)$ and proposal $q(\theta'|\theta)$, Barker's acceptance probability is:

$$\alpha_B(\theta, \theta') = \frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta) + \pi(\theta')q(\theta|\theta')} \tag{2.19}$$

While Barker's algorithm has lower efficiency than Metropolis-Hastings in the Peskun ordering, its acceptance probability has a crucial property: it can be expressed as a ratio of products. This enables the use of *Bernoulli factory* methods to implement the acceptance decision without explicitly computing α_B.

2.5.1.1 The Two-Coin Algorithm

The fundamental building block is the two-coin algorithm (Gonçalves; Łatuszyński; Roberts, 2017). Suppose we can factorize the acceptance odds as:

$$\frac{\pi(\theta')q(\theta|\theta')}{\pi(\theta)q(\theta'|\theta)} = \frac{c(\theta',\theta)p(\theta',\theta)}{c(\theta,\theta')p(\theta,\theta')} \tag{2.20}$$

where $c(\cdot,\cdot)$ is tractable and $p(\cdot,\cdot) \in [0,1]$ is a probability we can simulate.

Then, Barker's acceptance probability becomes:

$$\alpha_B(\theta,\theta') = \frac{c(\theta',\theta)p(\theta',\theta)}{c(\theta',\theta)p(\theta',\theta) + c(\theta,\theta')p(\theta,\theta')} \tag{2.21}$$

The two-coin algorithm generates this probability without computing it explicitly:

Algorithm 1: Two-coin algorithm Just the idea

1 Flip coins with probabilities $p(\theta',\theta)$ and $p(\theta,\theta')$;
2 **if** *both heads or both tails* **then**
3 | return to step 1;
4 **else if** *first heads and second tails* **then**
5 | accept (return 1);
6 **else if** *first tails and second heads* **then**
7 | reject (return 0);

This procedure produces an acceptance event with exactly probability $\alpha_B(\theta,\theta')$.

2.5.2 Divide-and-Conquer Bernoulli Factory (DCBF)

For models with n observations, the likelihood factorizes as $\pi(\theta|D) \propto \prod_{i=1}^n f_i(\theta)$. The vanilla two-coin algorithm requires flipping a coin with probability $p(\theta) = \prod_{i=1}^n p_i(\theta)$. As n grows, $p(\theta)$ shrinks exponentially, causing the expected number of loops (ENL) to grow as $O(e^{cn})$ for some constant $c > 0$. This exponential cost is prohibitive even for moderate n

2.5.2.1 The Coin Merge Procedure

The DCBF addresses this by hierarchically merging factor coins. For two factors with odds h_1, h_2 and corresponding probabilities $r_j = h_j/(h_j + g_j)$, we can merge them into a single coin with probability $r_0 = h_0/(h_0 + g_0)$ where $h_0 = h_1 h_2$.

Algorithm 2: Coin merge algorithm Just the idea (again)

```
Flip coins with probabilities r_1 and r_2;
if both return the same value (both 0 or both 1) then
    return that value;
else
    return to step 1;
end
```

2.5.2.2 Hierarchical Factorization

Given n factors, we organize them in a balanced binary tree of depth $\ell = \lceil \log_2 n \rceil$.

At the leaves, we place individual factors (or small batches). At each internal node, we merge the coins from its two children.

Algorithm 3: Divide-and-Conquer Bernoulli Factory

```
function FlipDCBF({f_1, ..., f_n}, level k)
    if n = 1 then
        return Flip2Coin(f_1);
    end
    Split factors into {f_1, ..., f_{n/2}} and {f_{n/2+1}, ..., f_n};
    c_1 <- FlipDCBF(left half, k + 1);
    c_2 <- FlipDCBF(right half, k + 1);
    return MergeCoins(c_1, c_2);
end
```

2.5.2.3 Computational Complexity

(Stumpf-Fétizon; Gonçalves, 2025) prove that under regularity conditions, the expected cost of the DCBF scales as:

$$\mathbb{E}[\text{Cost}] = O(4^{\ell} \cdot \rho(n, \ell)) \tag{2.22}$$

where $\rho(n, \ell)$ is the cost of the two-coin algorithm at the leaves with batch size $n/2^{\ell}$.

For posterior concentration at rate $n^{-1/2}$ with optimal proposal scaling:

- If using likelihood ratio estimators: $\rho(n, \ell) = O(\sqrt{n/2^{\ell}})$
- Setting $4^{\ell} \propto n$ gives overall cost $O(n^{3/2})$

This is polynomial, compared to the exponential $O(e^{cn})$ cost of vanilla two-coin.

Moreover, this matches or improves upon pseudo-marginal algorithms, which typically require $O(n^2)$ cost.

2.5.3 Pseudo-marginal Metropolis-Hastings

2.6 Function Emulation Methods

2.6.1 Gaussian Processes

2.6.2 Generalized Additive Models

2.6.3 Shape-constrained Variants

3 Approximate inference with noisy-MCMC

Current approaches:

- Function emulation: (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021)

`Luiz:` (Han et al., 2023) (Carvalho; Ibrahim, 2021) and references therein

`Luiz:` Need to carefully read, cite and discuss (Park; Haran, 2020). Do their results hold for SCAM, for instance?

Our contributions in this area:

- Do MaL estimation and talk about GP and SCAM;
- Importance sampling to simplify and improve estimation of the NPP normalising constant (including control variates);
- Showcase on moderate size examples (including NIG linear regression as a baseline). Show that it works and then show how to break it

3.1 Emulating the normalizing constant

`Adame:` Here I'd like to present the formulation of the problem, what exactly I mean by emulating in this case.

3.1.1 Consequences

`Adame:` Here I'd like to talk about how appoximations w/o uncertainty quantification might lead to unstrustworthy results

3.1.2 Introducing shape-constraints

`Adame:` Show how to plug our scenario into these models

3.2 Current approaches for the NPP

`Adame:` Discuss (Han et al., 2023) and (Carvalho; Ibrahim, 2021) - and any other state-of-art

3.3 Doubly Intractable Settings

Adame: Discuss (Park; Haran, 2020)

Doubly intractable distributions arise in Bayesian inference when the likelihood takes the form

$$\pi(\theta \mid x) \propto p(\theta)\frac{h(x \mid \theta)}{Z(\theta)}, \tag{3.1}$$

where the normalizing constant $Z(\theta)$ depends on the parameter θ and cannot be evaluated pointwise, as discussed in detail on seciton 2.3. This setting is common in models such as exponential random graph models, Markov point processes, and, more generally, in likelihoods derived from Gibbs measures. In the context of Normalized Power Priors (NPP), this issue naturally appears due to the presence of parameter-dependent normalizing constants, making standard Bayesian computation challenging.

A number of Markov chain Monte Carlo (MCMC) methods have been proposed to address this problem, including auxiliary variable approaches such as the exchange algorithm and the double Metropolis–Hastings (DMH) algorithm. While these methods can target the exact posterior distribution, they typically require simulating auxiliary variables at each iteration, which becomes computationally prohibitive in high-dimensional or complex models.

An alternative strategy is proposed by Park and Haran (2020), who introduce a function emulation approach to approximate the intractable normalizing constant. Their method replaces repeated Monte Carlo estimation of $Z(\theta)$ with a two-stage surrogate model. First, importance sampling is used to obtain noisy estimates of $Z(\theta)$ at a finite set of parameter values. Then, a Gaussian process (GP) is fitted to the log-normalizing function, providing a fast interpolator that can be evaluated within an MCMC algorithm. This leads to two variants: one that emulates only the normalizing function (NormEm), and another that emulates the entire likelihood (LikEm).

The main advantage of this approach is computational efficiency. By shifting the costly Monte Carlo approximations to a precomputation stage, the resulting MCMC algorithm avoids repeated simulation of auxiliary variables, yielding substantial speed-ups compared to exact or pseudo-marginal methods. Moreover, the approach is naturally parallelizable and extends beyond doubly intractable settings to more general problems involving expensive likelihood evaluations.

However, this efficiency comes at the cost of asymptotic exactness. The resulting Markov chain targets an approximate posterior distribution, with an error that depends on both the number of importance samples and the number of design points used to train the Gaussian process. While theoretical bounds on the total variation distance are provided, they do not yield explicit rates, making it difficult to calibrate the approximation in practice.

Several additional limitations remain. First, the method relies on importance sampling estimates, which can exhibit high variance in moderate to high-dimensional settings, potentially degrading the quality of the surrogate. Second, the Gaussian process approximation assumes a smooth structure for the log-normalizing function and typically ignores heteroskedasticity induced by Monte Carlo noise. Third, the selection of design points (particles) is critical but performed using heuristic procedures such as short runs of DMH or approximate Bayesian computation, without a principled adaptive strategy. Finally, the approach does not scale well with the dimension of the parameter space due to the cubic complexity of Gaussian process regression and the increasing difficulty of covering the parameter space.

These limitations suggest several directions for future research. In particular, combining surrogate-based approximations with exact correction mechanisms, such as pseudo-marginal or delayed-acceptance schemes, could potentially recover asymptotic exactness while retaining computational gains. Additionally, replacing importance sampling with more robust estimators, and incorporating adaptive or local surrogate models, may improve both stability and scalability. Such developments are particularly relevant in the context of NPP models, where efficient and reliable inference remains an open challenge.

4 Exact inference for the Normalized Power Prior

In this section, we develop exact Markov Chain Monte Carlo methods for inference with the normalized power prior (NPP). The key challenge lies in handling the intractable normalizing constant, which we address through Russian-roulette estimators that preserve the exactness of our sampling scheme.

4.1 Barker's Method via Poisson Estimator

For the NPP, the joint posterior is:

$$\pi(\theta, \eta | D, D_0) \propto \frac{1}{c_0(\eta)} L(D|\theta) L_0(D_0|\theta)^\eta \pi_0(\theta) \pi_A(\eta) \tag{4.1}$$

The challenging term is the normalizing constant:

$$c_0(\eta) = \int_\Theta L_0(D_0|\theta)^\eta \pi_0(\theta) \, \mathrm{d}\theta \tag{4.2}$$

4.1.1 Updating θ given η

`Adame:` Here I suggest substituting to regular MH

For updating θ, the full conditional is:

$$\pi(\theta | \eta, D, D_0) \propto L(D|\theta) L_0(D_0|\theta)^\eta \pi_0(\theta) \tag{4.3}$$

Since $L_0(D_0|\theta) = \prod_{i=1}^{N_0} f_i(\theta|D_{0i})$, we can use DCBF. For instance, when we have Gaussian data: $f_i(\theta|y_i) = \mathcal{N}(y_i|\theta, \Sigma)$.

The ratio form for Barker's algorithm is:

$$\frac{\pi(\theta'|\eta)}{\pi(\theta|\eta)} = \frac{L(D|\theta')}{L(D|\theta)} \cdot \frac{\pi_0(\theta')}{\pi_0(\theta)} \cdot \prod_{i=1}^{N_0} \left(\frac{f_i(\theta'|D_{0i})}{f_i(\theta|D_{0i})} \right)^\eta \tag{4.4}$$

The first two ratios are tractable. For the product term, we construct factor coins using likelihood ratios, which benefit from posterior concentration.

4.1.2 Updating η given θ

Updating η is more subtle because the full conditional involves $c_0(\eta)$:

$$\pi(\eta | \theta, D, D_0) \propto \frac{\pi_A(\eta)}{c_0(\eta)} L_0(D_0|\theta)^\eta \tag{4.5}$$

The ratio becomes:

$$\frac{\pi(\eta'|\theta)}{\pi(\eta|\theta)} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \cdot \frac{c_0(\eta)}{c_0(\eta')} \cdot L_0(D_0|\theta)^{\eta'-\eta} \tag{4.6}$$

The term $c_0(\eta)/c_0(\eta')$ is intractable but can be estimated using

$$\frac{\mathrm{d}}{\mathrm{d}\tau} \log c_0(\tau) = \mathbb{E}_{\theta \sim \pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta)\right], \tag{4.7}$$

where $\pi_\tau(\theta) \propto L_0(D_0 \mid \theta)^\tau \pi_0(\theta)$ is the power posterior at temperature τ.

Integrating this identity from η to η' yields the log-ratio:

$$\Delta = \log \frac{c_0(\eta')}{c_0(\eta)} = \int_\eta^{\eta'} \mathbb{E}_{\theta \sim \pi_\tau(\theta)} \left[\log L_0(D_0 \mid \theta)\right] \mathrm{d}\tau\,. \tag{4.8}$$

4.1.3 Practical implementation

In practice, we replace the integral Δ with a simple Monte Carlo estimation. This is a unbiased estimator for the actual integral:

$$\widehat{\Delta}_a(\eta, \eta') = \frac{(\eta' - \eta)}{K} \sum_{k=1}^{K} \mathbb{E}_{\theta \sim \pi_{\tau_k}(\theta)} \left[\log L_0(D_0 \mid \theta)\right], \text{ and so} \tag{4.9}$$

$$\widehat{\Delta}_b(\eta, \eta') = \frac{(\eta' - \eta)}{M \cdot K} \sum_{k=1}^{K} \sum_{m=1}^{M} \log L_0\Big(D_0 \mid \widetilde{\theta}_{m_k}\Big), \tag{4.10}$$

where $\tau_k \sim \text{Uniform}(\eta, \eta')$, and $\Big(\widetilde{\theta}_{m_k} \mid \tau_k\Big) \sim \pi_{\tau_k}(\theta)$.

Thus, for a single sample $\tau_\star \sim \text{Uniform}(\eta, \eta')$, and $\Big(\widetilde{\theta}_\star \mid \tau_\star\Big) \sim \pi_{\tau_\star}(\theta)$ we have:`Adame:` Because it doesn't depend on the variance

$$\widehat{\Delta}(\eta, \eta') = (\eta' - \eta) \log L_0\Big(D_0 \mid \widetilde{\theta}_\star\Big) \tag{4.11}$$

4.1.4 Producing an unbiased estimator via a Poisson estimator

Let π_τ denote the power posterior associated with the historical data. For $\tau \in [0, 1]$, define

$$\pi_\tau(\theta) = \frac{L_0(D_0 \mid \theta)^\tau \pi_0(\theta)}{c_0(\tau)}, \qquad c_0(\tau) = \int_\Theta L_0(D_0 \mid t)^\tau \, \mathrm{d}P_0(t). \tag{4.12}$$

For two values $\eta, \eta' \in [0, 1]$, define

$$\Delta(\eta, \eta') = \log \frac{c_0(\eta')}{c_0(\eta)}. \tag{4.13}$$

Under the usual regularity conditions allowing differentiation under the integral sign, the thermodynamic identity gives

$$\Delta(\eta, \eta') = \int_{\eta}^{\eta'} \mathbb{E}_{\theta \sim \pi_\tau} \left[\log L_0(D_0 \mid \theta)\right] \, \mathrm{d}\tau. \tag{4.14}$$

Equivalently, if $U \sim \mathrm{Unif}(0,1)$ and $\tau_{\eta,\eta'}(U) = \eta + U(\eta' - \eta)$, then

$$\Delta(\eta, \eta') = \mathbb{E}_U \left[(\eta' - \eta) \mathbb{E}_{\theta \sim \pi_{\tau_{\eta,\eta'}(U)}} \{\log L_0(D_0 \mid \theta)\} \right]. \tag{4.15}$$

This representation suggests a proposal-specific random variable. Let $\mathsf{P}_{\eta,\eta'}$ be the probability measure on $[0,1] \times \Theta$ obtained by first drawing $U \sim \mathrm{Unif}(0,1)$ and then drawing $\theta \sim \pi_{\tau_{\eta,\eta'}(U)}$. Equivalently,

$$\mathsf{P}_{\eta,\eta'}(\mathrm{d}u, \mathrm{d}\theta) = \mathrm{d}u \, \pi_{\tau_{\eta,\eta'}(u)}(\mathrm{d}\theta). \tag{4.16}$$

Define

$$Z_{\eta,\eta'}(U, \theta) = (\eta' - \eta) \log L_0(D_0 \mid \theta). \tag{4.17}$$

Then

$$\mathbb{E}_{\mathsf{P}_{\eta,\eta'}}[Z_{\eta,\eta'}] = \Delta(\eta, \eta'). \tag{4.18}$$

The objective is to construct an unbiased estimator of the ratio $c_0(\eta)/c_0(\eta')$. This ratio can be written as

$$\exp\{-\Delta(\eta, \eta')\} = \frac{c_0(\eta)}{c_0(\eta')}. \tag{4.19}$$

Assume that there exist deterministic constants $a_{\eta,\eta'}$ and $b_{\eta,\eta'}$ such that

$$a_{\eta,\eta'} \le Z_{\eta,\eta'}(U, \theta) \le b_{\eta,\eta'} \qquad \mathsf{P}_{\eta,\eta'}\text{-almost surely.} \tag{4.20}$$

Set

$$c_{\eta,\eta'} = b_{\eta,\eta'}, \qquad \lambda_{\eta,\eta'} = b_{\eta,\eta'} - a_{\eta,\eta'}. \tag{4.21}$$

Draw $K \sim \mathrm{Poisson}(\lambda_{\eta,\eta'})$. Conditionally on K, draw

$$(U_j, \theta_j)_{j=1}^K \overset{\text{iid}}{\sim} \mathsf{P}_{\eta,\eta'}. \tag{4.22}$$

The Poisson estimator is

$$\widehat{I}(\eta, \eta') = \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\} \prod_{j=1}^{K} \frac{c_{\eta,\eta'} - Z_{\eta,\eta'}(U_j, \theta_j)}{\lambda_{\eta,\eta'}}. \tag{4.23}$$

By construction, each factor satisfies

$$0 \le \frac{c_{\eta,\eta'} - Z_{\eta,\eta'}(U_j, \theta_j)}{\lambda_{\eta,\eta'}} \le 1. \tag{4.24}$$

Therefore $\widehat{I}(\eta, \eta')$ is non-negative. Using the exponential formula for a Poisson random variable, we obtain

$$\mathbb{E}[\widehat{I}(\eta, \eta')] = \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\}\mathbb{E}\left[\left\{\mathbb{E}_{\mathsf{P}_{\eta,\eta'}}\left[\frac{c_{\eta,\eta'} - Z_{\eta,\eta'}}{\lambda_{\eta,\eta'}}\right]\right\}^K\right] \tag{4.25}$$

$$= \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\}\exp\left\{\lambda_{\eta,\eta'}\left(\frac{c_{\eta,\eta'} - \Delta(\eta,\eta')}{\lambda_{\eta,\eta'}} - 1\right)\right\} \tag{4.26}$$

$$= \exp\{-\Delta(\eta, \eta')\}. \tag{4.27}$$

Thus, $\widehat{I}(\eta, \eta')$ is an unbiased estimator of the normalizing-constant ratio $c_0(\eta)/c_0(\eta')$.

The efficiency of the estimator depends strongly on the envelope width $\lambda_{\eta,\eta'} = b_{\eta,\eta'} - a_{\eta,\eta'}$. Large envelopes lead to large Poisson counts and high estimator variability. Therefore, the practical usefulness of the construction depends on obtaining deterministic but sufficiently tight proposal-specific bounds for $Z_{\eta,\eta'}$.

It is important to distinguish this unbiased estimator from an exact Barker accept–reject mechanism. Although $\widehat{I}(\eta, \eta')$ is unbiased for $c_0(\eta)/c_0(\eta')$, substituting a noisy estimate of the likelihood ratio directly into the Barker probability does not generally produce an exact Barker transition. Exact Barker implementations require simulating the accept–reject event itself with the correct marginal probability, typically through a Bernoulli-factory or two-coin construction.

4.1.5 Local envelopes for the Poisson estimator

The preceding construction does not require a single global envelope that is valid uniformly over all possible proposals. Instead, for each proposed move $\eta \to \eta'$, it is enough to obtain deterministic bounds for the proposal-specific random variable

$$Z_{\eta,\eta'}(U, \theta) = (\eta' - \eta)\log L_0(D_0 \mid \theta), \qquad (U, \theta) \sim \mathsf{P}_{\eta,\eta'}. \tag{4.28}$$

Suppose that the historical log-likelihood satisfies

$$\ell_{\min} \le \log L_0(D_0 \mid \theta) \le \ell_{\max} \qquad \text{for all } \theta \in \Theta. \tag{4.29}$$

Then, for a fixed proposal $\eta \to \eta'$, define

$$a_{\eta,\eta'} = \min\{(\eta' - \eta)\ell_{\min}, (\eta' - \eta)\ell_{\max}\}. \tag{4.30}$$

Similarly, define

$$b_{\eta,\eta'} = \max\{(\eta' - \eta)\ell_{\min}, (\eta' - \eta)\ell_{\max}\}. \tag{4.31}$$

The resulting envelope width is

$$b_{\eta,\eta'} - a_{\eta,\eta'} = |\eta' - \eta|(\ell_{\max} - \ell_{\min}). \tag{4.32}$$

Hence a natural local choice is

$$c_{\eta,\eta'} = b_{\eta,\eta'}, \qquad \lambda_{\eta,\eta'} = |\eta' - \eta|(\ell_{\max} - \ell_{\min}). \tag{4.33}$$

This choice makes the expected number of Poisson factors proportional to the proposal distance $|\eta' - \eta|$. Consequently, local random-walk proposals in η lead to smaller expected computational cost than large independent proposals.

For unbounded parameter spaces, such as Gaussian models with $\Theta = \mathbb{R}^d$, the lower bound $\ell_{\min}$ is typically $-\infty$. In that case, the above exact Poisson construction is not directly applicable without additional structure. One possible remedy is to work with a compactly supported prior. For example, define

$$\pi_{0,B}(\theta) \propto \pi_0(\theta)\mathbf{1}_{\Theta_B}(\theta), \tag{4.34}$$

where $\Theta_B \subset \Theta$ is compact. Then deterministic bounds for $\log L_0(D_0 \mid \theta)$ can be derived on Θ_B. In the Gaussian historical likelihood with known covariance, the log-likelihood is a concave quadratic function of θ. Its maximum is attained at the historical maximum likelihood estimator. Its minimum over a compact box is attained on the boundary. These deterministic bounds yield a valid local envelope for the Poisson estimator.

4.1.6 Divide-and-conquer factorization

The computational difficulty of the Poisson estimator increases with the variability of the historical log-likelihood. Since the historical likelihood factorizes over observations, we can write

$$\log L_0(D_0 \mid \theta) = \sum_{i=1}^{N_0} \log L_i(d_{0i} \mid \theta). \tag{4.35}$$

The width of a global envelope for $Z_{\eta,\eta'} = (\eta' - \eta)\log L_0(D_0 \mid \theta)$ typically grows with N_0. This motivates a divide-and-conquer construction.

Let $D_{0,1}, \ldots, D_{0,G}$ be a partition of the historical data. Let $\mathcal{I}_g$ denote the set of observation indices in block g. Define the block likelihood by

$$L_{0,g}(D_{0,g} \mid \theta) = \prod_{i \in \mathcal{I}_g} L_i(d_{0i} \mid \theta). \tag{4.36}$$

Then the full historical likelihood satisfies

$$L_0(D_0 \mid \theta) = \prod_{g=1}^{G} L_{0,g}(D_{0,g} \mid \theta). \tag{4.37}$$

Equivalently, the full historical log-likelihood satisfies

$$\log L_0(D_0 \mid \theta) = \sum_{g=1}^{G} \log L_{0,g}(D_{0,g} \mid \theta). \tag{4.38}$$

For each block g, define the blockwise random variable

$$Z_{g,\eta,\eta'}(U,\theta) = (\eta' - \eta)\log L_{0,g}(D_{0,g} \mid \theta). \tag{4.39}$$

Assume that deterministic bounds $a_{g,\eta,\eta'}$ and $b_{g,\eta,\eta'}$ are available such that

$$a_{g,\eta,\eta'} \leq Z_{g,\eta,\eta'}(U,\theta) \leq b_{g,\eta,\eta'} \qquad \mathsf{P}_{\eta,\eta'}\text{-almost surely.} \tag{4.40}$$

Set

$$c_{g,\eta,\eta'} = b_{g,\eta,\eta'}, \qquad \lambda_{g,\eta,\eta'} = b_{g,\eta,\eta'} - a_{g,\eta,\eta'}. \tag{4.41}$$

A blockwise Poisson estimator can then be constructed as

$$\widehat{I}_g(\eta,\eta') = \exp\{\lambda_{g,\eta,\eta'} - c_{g,\eta,\eta'}\} \prod_{j=1}^{K_g} \frac{c_{g,\eta,\eta'} - Z_{g,\eta,\eta'}(U_{g,j},\theta_{g,j})}{\lambda_{g,\eta,\eta'}}, \tag{4.42}$$

where $K_g \sim \text{Poisson}(\lambda_{g,\eta,\eta'})$. The variables $(U_{g,j},\theta_{g,j})$ are sampled independently from the same proposal-specific measure $\mathsf{P}_{\eta,\eta'}$. The blockwise estimator satisfies

$$\mathbb{E}[\widehat{I}_g(\eta,\eta')] = \exp\{-\Delta_g(\eta,\eta')\}, \tag{4.43}$$

where

$$\Delta_g(\eta,\eta') = \mathbb{E}_{\mathsf{P}_{\eta,\eta'}}\left[Z_{g,\eta,\eta'}(U,\theta)\right]. \tag{4.44}$$

If the block estimators are independent, the divide-and-conquer product estimator is

$$\widehat{I}_{\text{dc}}(\eta,\eta') = \prod_{g=1}^{G} \widehat{I}_g(\eta,\eta'). \tag{4.45}$$

By independence, its expectation is

$$\begin{aligned}
\mathbb{E}[\widehat{I}_{\text{dc}}(\eta,\eta')] &= \prod_{g=1}^{G} \mathbb{E}[\widehat{I}_g(\eta,\eta')] && (4.46)\\
&= \prod_{g=1}^{G} \exp\{-\Delta_g(\eta,\eta')\} && (4.47)\\
&= \exp\left\{-\sum_{g=1}^{G} \Delta_g(\eta,\eta')\right\}. && (4.48)
\end{aligned}$$

Since

$$\sum_{g=1}^{G} \Delta_g(\eta,\eta') = \Delta(\eta,\eta'), \tag{4.49}$$

we obtain

$$\mathbb{E}[\widehat{I}_{\text{dc}}(\eta,\eta')] = \exp\{-\Delta(\eta,\eta')\}. \tag{4.50}$$

Thus, the divide-and-conquer product estimator is unbiased for $c_0(\eta)/c_0(\eta')$.

The main advantage of this construction is variance and cost control. Instead of using one global envelope with width $\lambda_{\eta,\eta'}$, the method uses several smaller envelopes with widths $\lambda_{g,\eta,\eta'}$. The expected number of Poisson factors is

$$\sum_{g=1}^{G} \lambda_{g,\eta,\eta'}. \tag{4.51}$$

When the blockwise envelopes are much tighter than the full-data envelope, the product construction can be substantially more stable than the global estimator.

This product estimator should not be confused with a genuine recursive divide-and-conquer Bernoulli factory. The estimator $\widehat{I}_{\mathrm{dc}}(\eta, \eta')$ is an unbiased estimator of a normalizing-constant ratio. However, using it as a noisy plug-in inside the Barker probability does not generally yield an exact Barker transition. A true DCBF Barker transition recursively constructs the accept–reject coin itself. At the leaves, one simulates smaller Bernoulli coins associated with subsets of likelihood factors. At internal nodes, these coins are merged recursively until a single coin with the Barker acceptance probability is obtained at the root. Therefore, the recursive DCBF is a coin construction, not merely a product of random ratio estimates.

For the normalized power prior, the joint posterior can be written as

$$\pi(\theta, \eta \mid D, D_0) \propto \frac{L(D \mid \theta) \prod_{i=1}^{N_0} L_i(d_{0i} \mid \theta)^{\eta} \pi_0(\theta) \pi_A(\eta)}{c_0(\eta)}. \tag{4.52}$$

The normalizing constant depends on η, not on θ. It is given by

$$c_0(\eta) = \int_{\Theta} \prod_{i=1}^{N_0} L_i(d_{0i} \mid t)^{\eta} \, \mathrm{d}P_0(t). \tag{4.53}$$

Therefore, the correct factorization is through the historical likelihood terms $L_i(d_{0i} \mid \theta)$ and through proposal-specific envelopes for their log-contributions. In particular, the denominator should be written as $c_0(\eta)$ rather than $c_0(\theta)$.

In the experiments, we distinguish between two objects. The first object is a divide-and-conquer product Poisson estimator, which is an unbiased estimator of a normalizing-constant ratio under valid deterministic envelopes. The second object is a genuine divide-and-conquer Barker/Bernoulli-factory transition, which must simulate the accept–reject event itself. This distinction is essential because the first object is useful for diagnostics and pseudo-marginal comparisons, while the second object is required for exact Barker sampling.

4.2 Pseudo-marginal via Importance Sampling

An alternative to constructing an unbiased estimator of the ratio $c_0(\eta)/c_0(\eta')$ is to directly estimate the normalizing constant $c_0(\eta)$ itself and form the ratio from two independent estimates.

The simplest such approach is the *prior Monte Carlo estimator*, which exploits the representation

$$c_0(\eta) = \mathbb{E}_{\theta \sim \pi_0}[L_0(\theta \mid D_0)^\eta].$$

Samples drawn directly from the prior π_0 are cheap to obtain and require no gradient information or auxiliary computations. The main limitation is that the prior and the power posterior $\pi_\eta(\theta) \propto L_0(\theta \mid D_0)^\eta \pi_0(\theta)$ can be poorly aligned, particularly when the historical data are informative, leading to high variance in the importance weights and an inefficient estimator.

A more principled alternative is *importance sampling* centered at the maximum likelihood estimate. Writing the normalizing constant as

$$c_0(\eta) = \int_{\theta \in \Theta} L_0(\theta \mid D_0)^\eta \pi_0(\theta)\,\mathrm{d}\theta = \mathbb{E}_{\theta \sim \pi_{\mathrm{MLE}}}\left[L_0(\theta \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{\mathrm{MLE}}(\theta)}\right],$$

one draws samples from a proposal π_{MLE} centered at the MLE and reweights by the ratio $\pi_0(\theta)/\pi_{\mathrm{MLE}}(\theta)$. The MLE-centered proposal concentrates mass where the likelihood is large, reducing variance relative to the prior estimator when the historical data are informative. The cost is the need to compute or approximate the MLE, which may itself require numerical optimization.

5 Experiments

`Adame:` Add the new estimators. Is it methodology? Because it is still an example.

5.1 Assessing exact and approximate samplers

A central distinction in this chapter is between asymptotic exactness and finite-time computational efficiency. An MCMC transition kernel may be π-invariant, and hence exact in the usual Markov chain sense, while still being practically inefficient if its accept-reject mechanism is driven by rare events or high-variance unbiased estimators. For this reason, we do not compare methods only at a fixed number of iterations.

Instead, we report the computational time required to reach a target effective sample size for η, together with distributional discrepancies from a reference posterior. Since η is one-dimensional, we use the empirical Wasserstein distance as the primary convergence diagnostic. KL and Jensen-Shannon divergences are reported only as secondary diagnostics, since they require density estimation.

For exact Poisson and Bernoulli-factory constructions, we additionally monitor the local envelope parameter λ, the realized Poisson count K, the variability of $\log \widehat{I}$, invalid factors, and, for DCBF, recursive agreement attempts. These diagnostics separate correctness of the invariant distribution from practical viability of the resulting algorithm.

5.1.1 Building an estimator

The most simple approach is the harmonic estimator, since

$$c_0(\eta) = \mathbb{E}_{\theta \sim \pi_0}[L_0(\theta \mid D_0)^\eta].$$

Then, we define δ_a as

$$\delta_a(\eta, \theta_1, \dots, \theta_K) = \frac{1}{K} \sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \tag{5.1}$$

And with this estimator we can then employ a Portkey Barker's (Vats et al., 2021) to sample from the power posterior.

This δ_a has a major two-sided feature, while it is easy to sample from the initial prior, its efficiency is low due to its large non-informativeness.

In order to improve the efficiency, we can produce an importance sampling estimator by concentrating the probability density around the $\hat{\theta}_{\text{MLE}}$, which gives us

$$c_0(\eta) = \int_{\theta\in\Theta} L_0(\theta \mid D_0)^\eta \pi_0(\theta)\,\mathrm{d}\theta = \mathbb{E}_{\theta\sim\pi_{\text{MLE}}}\left[L_0(\theta \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{\text{MLE}}(\theta)}\right].$$

We can then define δ_b as:

$$\delta_b(\eta, \theta_1, \ldots, \theta_K) = \frac{1}{K}\sum_{i=1}^{K} L_0(\theta_i \mid D_0)^\eta \frac{\pi_0(\theta_i)}{\pi_{\text{MLE}}(\theta_i)}. \tag{5.2}$$

`Adame:` Different compatibility scenarios.

5.1.2 The choice of λ and c for the Gaussian likelihood with known Σ.

Considering the functional form of our estimator $\widehat{\Delta}(\eta, \eta') = (\eta' - \eta)\log L_0\big(D_0 \mid \widetilde{\theta}\big)$ and that L_0 is the likelihood of $D_0 = \{d_{01}, \ldots, d_{0N}\}$ where each $d_{0j} \sim \mathcal{N}_d(\theta, \Sigma)$ for $j = 1, \ldots, N$ with known Σ, we can find that

$$\arg\max_{\theta\in\Theta} \log L_0(D_0 \mid \theta) = \hat{\theta}_{\text{MLE}} = \frac{1}{N}\sum_{j=1}^{N} d_{0j},$$

and, therefore, we have that

$$\max_{\theta\in\Theta} \log L_0(D_0 \mid \theta) = -\frac{Nd}{2}\log(2\pi) - \frac{N}{2}\log\det\Sigma - \frac{N}{2}\operatorname{tr}(\Sigma^{-1}S),$$

where $S = \frac{1}{N}\sum_{j=1}^{N}(x_j - \hat{\theta}_{\text{MLE}})(x_j - \hat{\theta}_{\text{MLE}})^\top$

5.1.3 The choice of λ and c for the exponential family

`Adame:` Produce analytical results for exponential family. I have it somewhere.

A canonical exponential family has the following density:

$$L(X \mid \theta) = b(X)\exp\Big\{\theta^\top t(X) - c(\theta)\Big\},\ \theta \in \Theta \subseteq \mathbb{R}^d, \tag{5.3}$$

where Θ is open and connected.

5.2 Using Exact Methods as Baseline

`Adame:` Here I'd like to use the best exact method to assess the quality of the approximate methods

5.3 Numeric Experiments

5.3.1 Convergence to the baseline

We evaluate the proposed estimators under a multivariate Gaussian model with known covariance ($p = 3$, $\Sigma = I_3$, $\Sigma_0 = 2I_3$), placing weakly informative Beta$(1, 1)$ priors on η and an initial $\mathcal{N}_3(0, \Sigma_0)$ prior on θ. Historical and current datasets each contain $N_0 = N = 50$ observations. Three discrepancy scenarios are constructed by shifting the historical data mean by $\delta \in \{0.1, 0.5, 1.0\}$ per coordinate, representing low, moderate, and high conflict between historical and current information.

Rather than running each method for a fixed number of iterations, we adopt a convergence-based protocol: each estimator is run in independent batches of four chains and 5 000 iterations until the two-sample Kolmogorov–Smirnov (KS) distance between its accumulated posterior draws for η and the Stan HMC baseline drops below 0.05. The number of chains and total wall-clock time required to reach this threshold serve as our primary efficiency metrics. The exact algorithm—which uses the closed-form normalising constant ratio—acts as a reference. We compare it against three approximate pseudo-marginal methods: the prior-importance estimator (δ_a), the MLE importance-sampling estimator (δ_b), and the power-posterior estimator.

	Small ($\delta = 0.1$)		Medium ($\delta = 0.5$)		Large ($\delta = 1.0$)	
Method	Chains	KS	Chains	KS	Chains	KS
Exact	16	0.029	8	0.033	4	0.010
Prior-importance	40	0.323	40	0.229	4	0.027
Importance	40	0.084	12	0.050	4	0.026
Power	8	0.017	4	0.033	4	0.021

Table 1 – Total chains required and final KS distance to the Stan HMC baseline. A dash indicates the method reached 10 batches (40 chains) without satisfying the convergence criterion.

5.3.2 Discussion of results

The results reveal a consistent pattern across discrepancy regimes. Under low discrepancy (Figure 1), the posterior for η spreads across a wide range, reflecting genuine uncertainty about how much weight to assign to the historical data. In this regime the estimators differ most sharply: the power estimator converges in just two batches (8 chains, KS$= 0.017$), the exact algorithm requires four batches (16 chains, KS$= 0.029$), whereas the importance estimator exhausts all ten batches without reaching the threshold (KS$= 0.084$) and the prior-importance estimator deviates substantially from the baseline (KS$= 0.323$). The inefficiency of δ_a here is expected: when the posterior for η is diffuse, prior samples

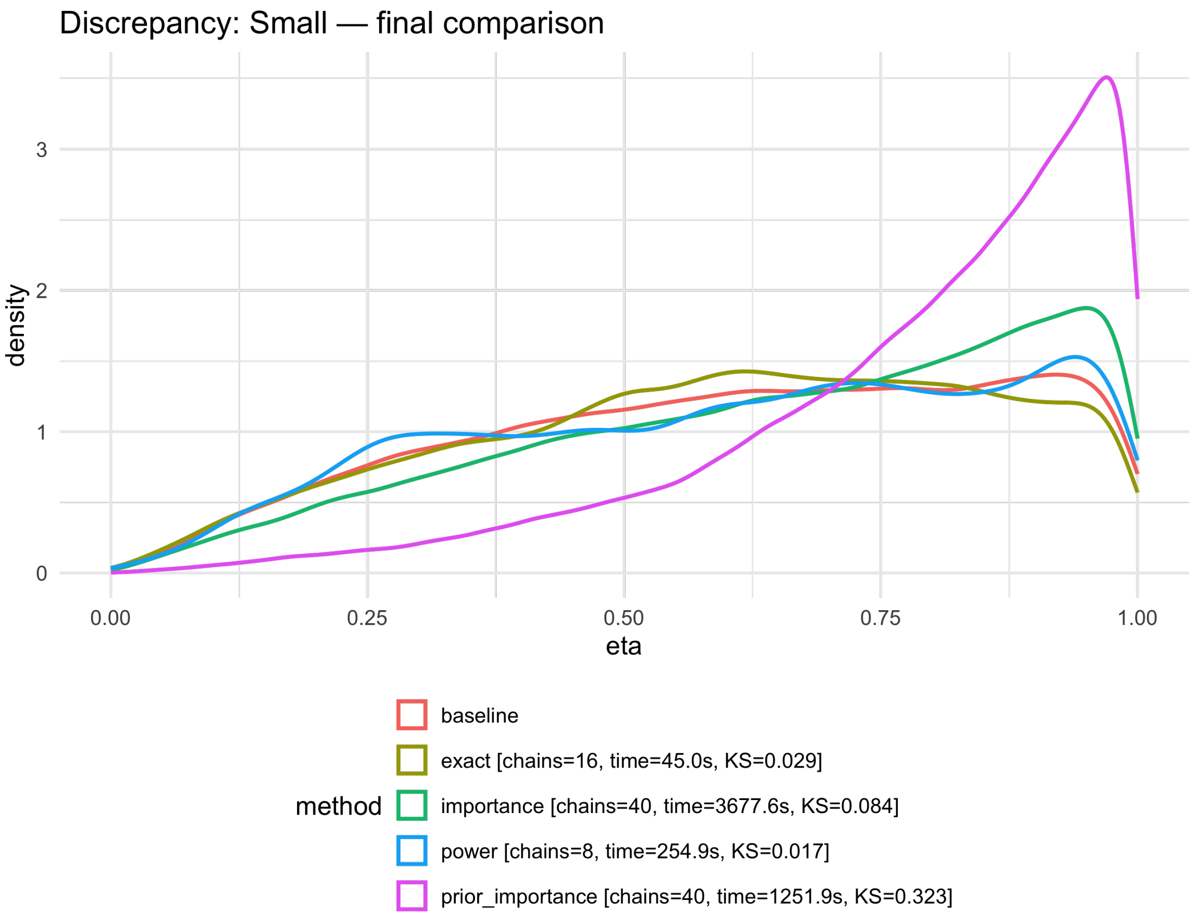

Figure 1 – Posterior densities for η under low discrepancy ($\delta = 0.1$). Each approximate estimator label reports the total number of chains run, wall-clock time, and final KS distance to the Stan baseline.

for θ are a poor proposal for the normalising-constant integral, yielding high-variance estimates of the log-ratio.

Under moderate discrepancy (Figure 2), the posterior shifts toward smaller values of η, indicating the model has begun to down-weight the historical data. The power estimator again converges in a single batch (4 chains), the exact algorithm in two, and the importance estimator in three. The prior-importance estimator still fails to reach the threshold, though its KS distance has improved relative to the low-discrepancy case.

When the two datasets are in strong conflict (Figure 3), the posterior concentrates sharply near $\eta \approx 0$. In this regime all four estimators converge in a single batch of four chains, and the KS distances are uniformly small. The sharp concentration of the posterior makes the normalising-constant ratio easy to estimate regardless of the proposal, erasing the differences observed at lower discrepancy levels.

Overall, the power estimator is the most efficient of the three approximate methods: it matches or beats the exact algorithm in chains required across all scenarios. The prior-importance estimator is consistently the weakest, particularly when the posterior is

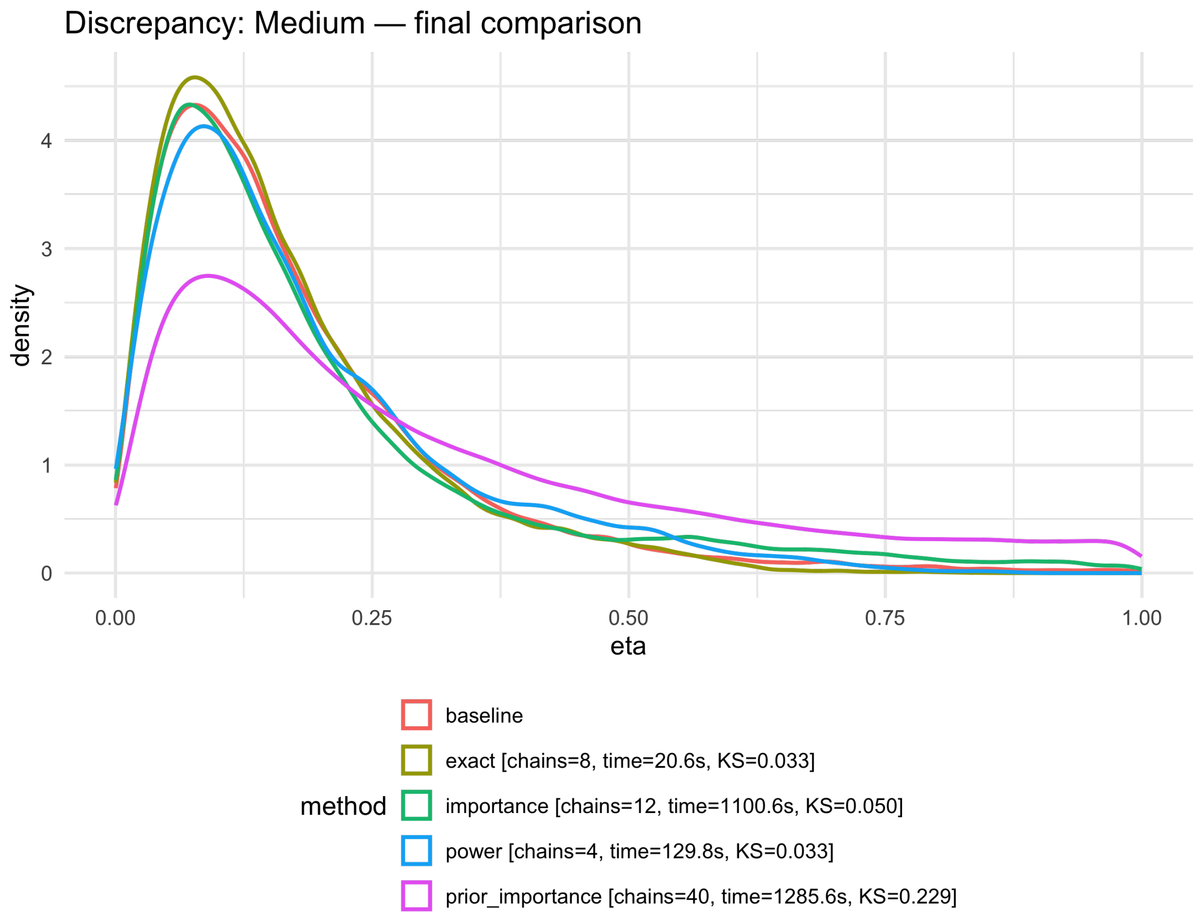

Figure 2 – Posterior densities for η under moderate discrepancy ($\delta = 0.5$).

diffuse. These findings suggest that the choice of estimator matters most in low-discrepancy settings, where the posterior for η is broad and the normalising constant must be estimated accurately over a wide range of values.

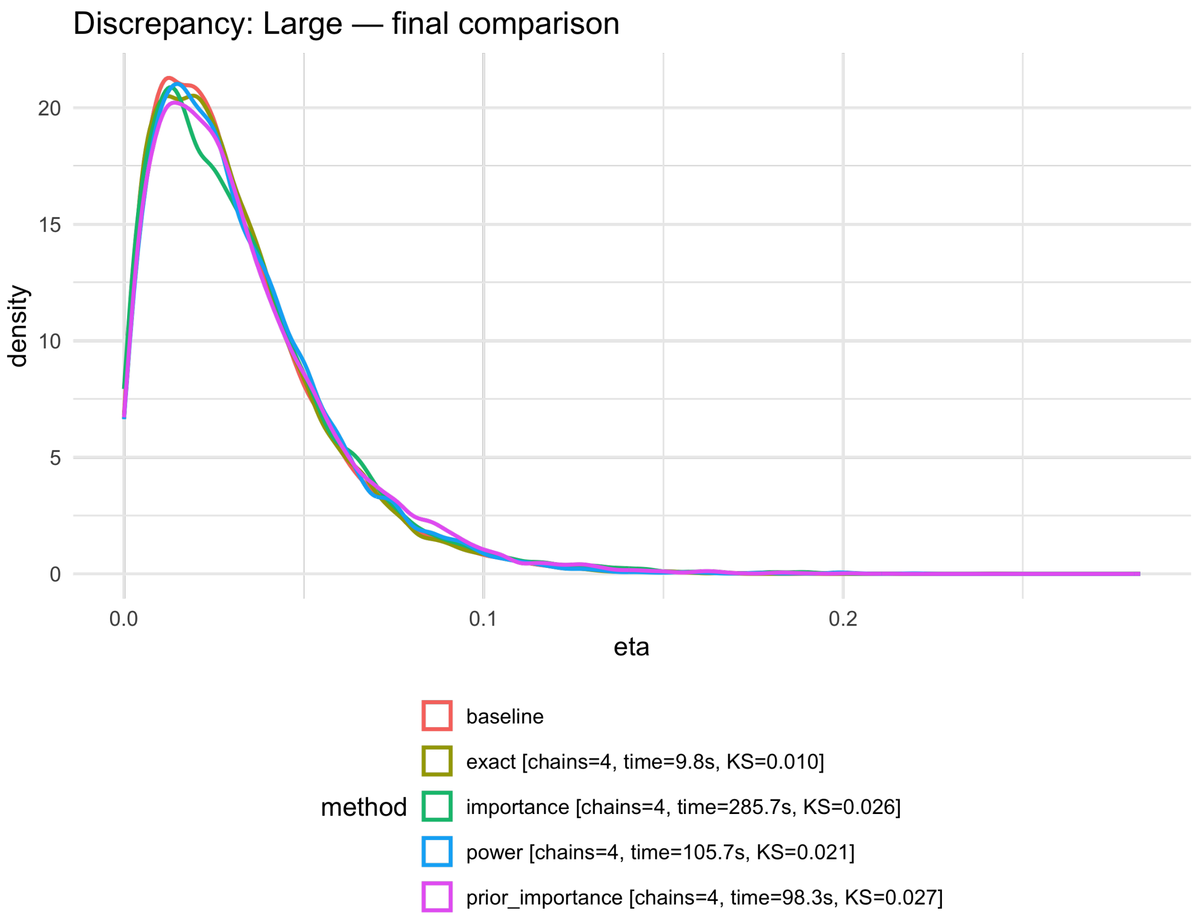

Figure 3 – Posterior densities for η under high discrepancy ($\delta = 1.0$). Under strong conflict between datasets the posterior concentrates near zero and all methods converge in a single batch.

6 Discussion

Code Availability The code is available at:

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APPENDIX A – Proofs

A.1 Proof that ?? is unbiased

Proof. The result is immediate, we first take the expected value with respect to P that leaves us with the actual Δ inside the expected value.

$$\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\dots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] = \mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\dots,(\tau_\star,\theta_\star)_K}\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\widehat{\Delta}(\tau_{\star j}))}{\lambda}\right], \tag{A.1}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\prod_{j=1}^{K}\frac{(c-\Delta)}{\lambda}\right], \tag{A.2}$$

$$= \mathbb{E}_K\left[\exp\{(\lambda - c)\}\left(\frac{(c-\Delta)}{\lambda}\right)^K\right], \tag{A.3}$$

then, we take the expected value with respect to $K \sim \text{Poisson}(\lambda)$

$$\begin{aligned}
\mathbb{E}_{K,(\tau_\star,\theta_\star)_1,\dots,(\tau_\star,\theta_\star)_K}\left[\widehat{I}\right] &= \sum_{i=0}^{\infty}\mathbb{P}[K=i]\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= \sum_{i=0}^{\infty}\frac{e^{-\lambda}\lambda^i}{i!}\exp\{\lambda - c\}\left(\frac{(c-\Delta)}{\lambda}\right)^i, \\
&= e^{-c}\sum_{i=0}^{\infty}\frac{(c-\Delta)}{i!} = e^{-c}e^{c-\Delta} = e^{-\Delta}.
\end{aligned}$$

$\square$

APPENDIX B – Formulations

B.1 Gaussian likelihood with known variance

Consider $d_{0j} \sim \mathcal{N}(\theta, \sigma^2)$ with known variance σ^2 for all $j \in 0, \ldots, p$, and let P_0 be the measure induced by the prior $\pi_0(\theta) \propto 1$.

Therefore, let $\bar{d}_{0p} = \sum_{j=1}^{p} d_{0j}/p$ be the mean of the historical data D_0 and $S^2 = \sum_{j=1}^{p}(d_{0j} - \bar{d}_{0p})^2$ the sample variance. Then, we have

$$
\begin{aligned}
c_0(\eta) &= \int_{\Theta} L(D_0 \mid t)^{\eta} \,\mathrm{d}P_0(t), \\
&= \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}\left(\sum_{j=1}^{p}(d_{0j} - t)^2\right)\right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{\eta}{2\sigma^2}\left(\sum_{j=1}^{p}((d_{0j} - \bar{d}_{0p}) + (\bar{d}_{0p} - t))^2\right)\right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{\eta}{2\sigma^2}\left(S^2 + p(t - \bar{d}_{0p})^2 + \cancelto{0}{\sum_{j=1}^{p} 2(d_{0j} - \bar{d}_{0p})(t - \bar{d}_{0p})}\right)\right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\} \int_{-\infty}^{\infty} \exp\left\{-\frac{\eta}{2\sigma^2}p(t - \bar{d}_{0p})^2\right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\} \sqrt{\frac{2\pi\sigma^2}{p\eta}} = \frac{1}{(2\pi\sigma^2)^{(\eta p-1)/2}\sqrt{p\eta}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\}.
\end{aligned}
$$

Computing the logarithm of the normalizing constant, we have

$$
\log c_0(\eta) = -\frac{(\eta p - 1)}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2}\log(p\eta) - \frac{\eta}{2\sigma^2}S^2.
$$

Therefore, the derivative of the logarithm of the normalizing constant is given by

$$
\frac{\mathrm{d}\log c_0(\eta)}{\mathrm{d}\eta} = -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{S^2}{2\sigma^2} - \frac{1}{2\eta}.
$$

On the other hand, we can show that $\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))]$ matches the previous expression. In this case, we have that $\theta \mid D_0 \sim \mathcal{N}(\bar{d}_{0p}, \sigma^2/p\eta)$, so we can compute the expectation as follows:

$$
\begin{aligned}
\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))] &= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \mathbb{E}_{\pi_\eta}\left[\frac{1}{2\sigma^2}\left(\sum_{j=1}^{p}(d_{0j}-\theta)^2\right)\right], \\
&= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\mathbb{E}_{\pi_\eta}\left[(d_{0j}-\theta)^2\right], \\
&= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\Big((d_{0j}-\mathbb{E}_{\pi_\eta}[\theta])^2 + \mathbb{V}_{\pi_\eta}[\theta]\Big), \\
&= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\left((d_{0j}-\bar{d}_{0p})^2 + \frac{\sigma^2}{p\eta}\right), \\
&= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}(d_{0j}-\bar{d}_{0p})^2 - \frac{p}{2\sigma^2}\frac{\sigma^2}{p\eta}, \\
&= -\frac{p}{2}\log\Big(2\pi\sigma^2\Big) - \frac{S^2}{2\sigma^2} - \frac{1}{2\eta}.
\end{aligned}
$$

B.1.1 Reformulation from Importance Sampling

One might be interested in using importance sampling to estimate the expectation in the last expression. In this case, we can rewrite the expectation as

$$
\begin{aligned}
c_0'(\eta) &= c_0(\eta)\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))] \\
&= c_0(\eta)\mathbb{E}_{P_0}\left[\log(L(D_0 \mid \theta))\frac{\pi_\eta(\theta)}{\pi_0(\theta)}\right] \\
&= c_0(\eta)\mathbb{E}_{P_0}\left[\log(L(D_0 \mid \theta))\frac{L_0(D_0 \mid \theta)^\eta}{c_0(\eta)}\right] = \mathbb{E}_{P_0}\left[\log(L(D_0 \mid \theta))L(D_0 \mid \theta)^\eta\right].
\end{aligned}
$$

B.1.2 Adding a Gaussian Initial Prior

Consider again $d_{0j} \sim \mathcal{N}(\theta, \sigma^2)$ with known variance σ^2 for all $j \in 0, \dots, p$, but now let $P_0 \equiv \mathcal{N}(0, \sigma_0^2)$, where σ_0^2 is a known initial prior variance.

Again, let $\bar{d}_{0p} = \sum_{j=1}^{p} d_{0j}/p$ be the mean of the historical data D_0 and $S^2 = \sum_{j=1}^{p}(d_{0j} - \bar{d}_{0p})^2$ the sample variance. Starting, we get:

$$
c_0(\eta) = \frac{1}{(2\pi\sigma^2)^{\eta p/2}}\exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\}\frac{1}{\sqrt{2\pi\sigma_0^2}}\int_{-\infty}^{\infty}\exp\left\{-\frac{\eta}{2\sigma^2}p(t-\bar{d}_{0p})^2 - \frac{1}{2\sigma_0^2}t^2\right\}\mathrm{d}t.
$$

Expanding $-\frac{\eta p}{2\sigma^2}(\bar{d}_{0p} - t)^2 - \frac{t^2}{2\sigma_0^2}$, we get $(\bar{d}_{0p} - t)^2 = t^2 - 2t\bar{d}_{0p} + \bar{d}_{0p}^2$. And, thus, we have that

$$
\begin{aligned}
-\frac{\eta p}{2\sigma^2}(\bar{d}_{0p}-t)^2-\frac{t^2}{2\sigma_0^2} &= -\frac{\eta p}{2\sigma^2}\left(t^2-2t\bar{d}_{0p}+\bar{d}_{0p}^{\,2}\right)-\frac{t^2}{2\sigma_0^2}, \\
&= -\frac{1}{2}\left(t^2\left(\frac{\eta p}{\sigma^2}+\frac{1}{\sigma_0^2}\right)-2t\bar{d}_{0p}\frac{\eta p}{\sigma^2}+\bar{d}_{0p}^{\,2}\frac{\eta p}{\sigma^2}\right).
\end{aligned}
$$

Define $A=\frac{\eta p}{\sigma^2}+\frac{1}{\sigma_0^2}$ and $B=\frac{\eta p}{\sigma^2}\bar{d}_{0p}$. We can now complete the square. Once we have that $At^2-2Bt=A\left(t-\frac{B}{A}\right)^2-\frac{B^2}{A}$, it follows that $-\frac{1}{2}(At^2-2Bt)=-\frac{A}{2}\left(t-\frac{B}{A}\right)^2+\frac{B^2}{2A}$. Which gives us

$$
-\frac{1}{2}\left(At^2-2Bt+\bar{d}_{0p}^{\,2}\frac{\eta p}{\sigma^2}\right)=-\frac{A}{2}\left(t-\frac{B}{A}\right)^2+\frac{B^2}{2A}-\frac{\bar{d}_{0p}^{\,2}}{2}\frac{\eta p}{\sigma^2}.
$$

Therefore, we can rewrite the normalizing constant as

$$
\begin{aligned}
c_0(\eta) &= \frac{1}{(2\pi\sigma^2)^{\eta p/2}}\exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\}\frac{1}{\sqrt{2\pi\sigma_0^2}}\int_{-\infty}^{\infty}\exp\left\{-\frac{A}{2}\left(t-\frac{B}{A}\right)^2+\frac{B^2}{2A}-\frac{\bar{d}_{0p}^{\,2}}{2}\frac{\eta p}{\sigma^2}\right\}\mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}}\exp\left\{-\frac{\eta}{2\sigma^2}S^2-\frac{\eta p}{2\sigma^2}\bar{d}_{0p}^{\,2}\right\}\cdot\frac{1}{\sqrt{A\sigma_0^2}}\cdot\exp\left\{\frac{B^2}{2A}\right\}, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}}\exp\left\{-\frac{\eta}{2\sigma^2}\left(S^2-p\bar{d}_{0p}^{\,2}\right)+\frac{B^2}{2A}\right\}\cdot\frac{1}{\sqrt{A\sigma_0^2}}.
\end{aligned}
$$

Taking the logarithm, we have

$$
\log c_0(\eta)=-\frac{(\eta p)}{2}\log\left(2\pi\sigma^2\right)-\frac{\eta}{2\sigma^2}\left(S^2-p\bar{d}_{0p}^{\,2}\right)+\frac{B^2}{2A}-\frac{1}{2}\log\left(A\sigma_0^2\right).
$$

And, thus, we can compute the derivative of the logarithm of the normalizing constant term by term. The first two terms and the last one can be calculated right away:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{(\eta p)}{2}\log\left(2\pi\sigma^2\right)\right) &= -\frac{p}{2}\log\left(2\pi\sigma^2\right), \\
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{\eta}{2\sigma^2}\left(S^2-p\bar{d}_{0p}^{\,2}\right)\right) &= -\frac{1}{2\sigma^2}\left(S^2-p\bar{d}_{0p}^{\,2}\right),\ \text{and} \\
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{1}{2}\log\left(A\sigma_0^2\right)\right) &= -\frac{1}{2}\left(\frac{A'}{A}\right)=\frac{-p/\sigma^2}{2A}.
\end{aligned}
$$

The third term is given by

$$\frac{d}{d\eta}\left(\frac{B^2}{2A}\right) = \frac{2BB'A - B^2A'}{2A^2},$$

where $B' = \frac{d}{d\eta}\left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right) = \frac{p}{\sigma^2}\bar{d}_{0p}$ and $A' = \frac{d}{d\eta}\left(\frac{\eta p}{\sigma^2} + \frac{1}{\sigma_0^2}\right) = \frac{p}{\sigma^2}$, thus

$$\begin{aligned} 2BB' &= 2\cdot\left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right)\cdot\left(\frac{p}{\sigma^2}\bar{d}_{0p}\right) = 2\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2}, \\ B^2 &= \left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right)^2 = \eta^2\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2}, \end{aligned}$$

then

$$\frac{2BB'A - B^2A'}{2A^2} = \frac{1}{2A^2}\left(2\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2}A - \eta^2\frac{p^3}{\sigma^6}\bar{d}_{0p}^{\,2}\right) = \frac{\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2}}{A^2}\left(A - \frac{\eta p}{2\sigma^2}\right).$$

And now we can recover the $c_0'(\eta)$ expression as

$$c_0'(\eta) = c_0(\eta)(\log(c_0(\eta)))' \tag{B.1}$$

and compare it with the importance sampling reformulation.

B.2 Linear Regression with NIG priors

Suppose we observe $\boldsymbol{y}_0 = \boldsymbol{X}_0\boldsymbol{\beta} + \varepsilon_0,\ \boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \varepsilon$ with $\varepsilon_0 \sim \mathcal{N}(0, \sigma^2 I_{n_0}),\ \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ and unknown variance σ^2. Now let $P_0 \equiv \mathcal{N}(\boldsymbol{\mu}_0,\ \boldsymbol{V}_0)\cdot \text{Inv-Gamma}(a, b)$, where $\boldsymbol{\mu}_0, \boldsymbol{V}_0, a$ and b are known hyperparameters.

Assume that $\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite and let

$$\begin{aligned} \hat{\boldsymbol{\beta}}_0 &= (\boldsymbol{X}_0^\top\boldsymbol{X}_0)^{-1}\boldsymbol{X}_0^\top\boldsymbol{y}_0, \\ \hat{\boldsymbol{\beta}} &= (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y}, \\ S_0 &= (\boldsymbol{y}_0 - \boldsymbol{X}_0\hat{\boldsymbol{\beta}}_0)^\top(\boldsymbol{y}_0 - \boldsymbol{X}_0\hat{\boldsymbol{\beta}}_0), \\ S &= (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}). \end{aligned}$$

So the likelihood of the historical data D_0 is

$$L(D_0 \mid \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n_0/2}\exp\left\{-\frac{1}{2\sigma^2}\left[S_0 + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_0)^\top\boldsymbol{X}_0^\top\boldsymbol{X}_0(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_0)\right]\right\}$$

and the initial prior of $(\boldsymbol{\beta}, \sigma^2)$ is

$$\pi_0(\boldsymbol{\beta}, \sigma^2) \propto (\sigma^2)^{-a-p/2} \exp\left\{-\frac{1}{\sigma^2}\left[b + \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu}_0)^\top \boldsymbol{V}_0^{-1}(\boldsymbol{\beta} - \boldsymbol{\mu}_0)\right]\right\}.$$

Then

$$\begin{aligned} c_0(\eta) = &\int_0^\infty \int_{\mathbb{R}^p} \pi_0(\boldsymbol{u}, v) L(D_0 \mid \boldsymbol{u}, v)^\eta d\boldsymbol{u} dv, \\ &\propto \int_0^\infty \int_{\mathbb{R}^p} (2\pi)^{-n_0\eta/2} v^{-a-p/2-n_0\eta/2} \exp\left\{-\frac{1}{2v}\left[2H_0(\eta) + (\boldsymbol{u} - \tilde{\boldsymbol{\beta}}_0)^\top (\boldsymbol{V}_0^{-1} + \eta \boldsymbol{X}_0^\top \boldsymbol{X}_0)(\boldsymbol{u} - \tilde{\boldsymbol{\beta}}_0)\right]\right\} d \\ &\propto (2\pi)^{-n_0\eta/2} \Gamma(\nu_0) \mid \Lambda_0 \mid^{-1/2} H_0(\eta)^{-\nu_0}, \end{aligned}$$

where

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_0 &= (\Lambda_0)^{-1}(\boldsymbol{V}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{X}_0^\top \boldsymbol{y}_0 \eta), \\ \nu_0 &= \frac{n_0\eta}{2} + a - 1, \\ \Lambda_0 &= \boldsymbol{V}_0^{-1} + \boldsymbol{X}_0^\top \boldsymbol{X}_0 \eta, \\ H_0(\eta) &= b + \frac{\eta}{2}\left[S_0 + (\boldsymbol{\mu}_0 - \hat{\boldsymbol{\beta}}_0)^\top \boldsymbol{X}_0^\top \boldsymbol{X}_0 (\Lambda_0)^{-1} \boldsymbol{V}_0^{-1} (\boldsymbol{\mu}_0 - \hat{\boldsymbol{\beta}}_0)\right]. \end{aligned}$$

This leads to the following NPP:

$$\pi(\boldsymbol{\beta}, \sigma^2, \eta \mid D_0) \propto \frac{\pi_0(\eta) H_0(\eta)^{\nu_0} \mid \Lambda_0 \mid^{1/2}}{(\sigma^2)^{\frac{n+n_0\eta+p}{2}+a}\Gamma(\nu_0)} \exp\left\{-\frac{1}{2\sigma^2}\left[(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}_0)^\top \Lambda_0 (\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}_0) + 2H_0(\eta)\right]\right\}.$$

Including the likelihood we can derive the joint posterior distribution:

$$\pi(\boldsymbol{\beta}, \sigma^2, \eta \mid D, D_0) \propto \frac{\pi_0(\eta) H_0(\eta)^{\nu_0} \mid \Lambda_0 \mid^{1/2}}{(\sigma^2)^{\frac{n+n_0\eta+p}{2}+a}\Gamma(\nu_0)} \exp\left\{-\frac{1}{2\sigma^2}\left[(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}})^\top \Lambda (\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}) + 2H(\eta)\right]\right\},$$

where

$$\begin{aligned} \tilde{\boldsymbol{\beta}} &= \Lambda^{-1}(\boldsymbol{V}_0^{-1}\mu_0 + \boldsymbol{X}^\top \boldsymbol{y} + \boldsymbol{X}_0^\top \eta), \\ \Lambda &= \Lambda_0 + \boldsymbol{X}^\top \boldsymbol{X}, \\ H(\eta) &= H_0(\eta) + \frac{1}{2}[S + (\tilde{\boldsymbol{\beta}}_0 - \hat{\boldsymbol{\beta}})^\top \boldsymbol{X}^\top \boldsymbol{X} \Lambda^{-1} \Lambda_0 (\tilde{\boldsymbol{\beta}}_0 - \hat{\boldsymbol{\beta}})]. \end{aligned}$$

So we can express the marginal posterior distribution of η given (D_0, D) as

$$\pi(\eta \mid D_0, D) \propto \frac{\pi_0(\eta)\Gamma(n/2+\nu_0) \mid \Lambda_0 \mid^{1/2} H_0(\eta)^{\nu_0}}{\mid \Lambda \mid^{1/2} \Gamma(\nu_0) H(\eta)^{\nu_0+n/2}},$$

and the conditional posterior distribution of $\boldsymbol{\beta}$ given (η, D_0, D) follows the multivariate Student distribution with location parameter $\tilde{\boldsymbol{\beta}}$, shape matrix Σ, and degree of freedom ρ, where $\rho = n + 2\nu_0$ and $\Sigma = \frac{2H(\eta)}{n+2\nu_0}\Lambda^{-1}$.